Submodel Partitioning in Hierarchical Federated Learning: Algorithm Design and Analysis

Wenzhi Fang, Dong-Jun Han, and Christopher G. Brinton
Elmore Family School of Electrical and Computer Engineering, Purdue University
Email: {fang375, han762, cgb}@purdue.edu

***Abstract*—Hierarchical federated learning (HFL) is currently regarded as a promising variant of traditional cloud-based federated learning (FL) due to its advantages in efficiency and scalability. However, HFL still suffers from significant computation, communication, and storage, especially when training a large-scale model over resource-constrained devices. In this paper, we introduce *hierarchical independent submodel training* (HIST), a new FL methodology that can address these issues in hierarchical settings. Our idea is to partition the global model into disjoint submodels in each round and distribute them across different cells, so that each cell is responsible for training only a specific partition of the full model. This enables each client to save computation/storage costs while alleviating the communication loads between clients and the edge server, and edge servers and the cloud server. We characterize the convergence behavior of HIST for non-convex loss functions with some mild assumptions, explore the performance-resource trade-off, and provide guidelines on designing the system parameters for HFL. Finally, through experiments, we demonstrate the effectiveness and efficiency of HIST.**

I. Introduction

The past decade has witnessed a huge breakthrough in various machine learning (ML) applications, including computer vision and natural language processing. As training data for these tasks are often collected by individual clients that are geographically separated, developing an efficient distributed training strategy has become increasingly important. In this context, federated learning (FL) is receiving significant attention nowadays as it enables distributed clients to collaboratively train a global model without any raw data exchange [1].

The traditional cloud-based FL requires all clients in the system to directly communicate with the central cloud server, resulting in communication and scalability issues. For this reason, hierarchical federated learning (HFL) was proposed [2]–[4], where the clients are clustered into different cells and each cell is served by an edge server. The introduction of edge servers in HFL reduces communication and scheduling complexity at the cloud server compared to the conventional single server FL, as the cloud server only needs to communicate with the edge servers in HFL. However, as the size of the model increases, HFL still suffers from significant (i) computation/storage at individual clients, (ii) communication between clients and the edge server, and (iii) communication between edge servers and the cloud server. These training requirements are the bottlenecks for practical deployment of HFL especially when resource-constrained clients aim to collaboratively train a large-scale model.

Contributions: In this paper, we propose HIST, a new FL methodology that integrates independent submodel training (IST) in hierarchical networks to address the aforementioned challenges. The core idea of HIST is to partition the global model into disjoint submodels in each global round and distribute them across different cells, so that devices in distinct cells are responsible for training different partitions of the full model. Such a submodel partitioning effectively reduces computation and storage at local clients, and also alleviates communication on both the links between clients and the edge server and between edge servers and the cloud server. Our main contributions are summarized as follows:

- We propose HIST, a hierarchical independent submodel training methodology that successfully reduces computation, communication, and storage costs during the training process of HFL.
- Under non-independent and identically distributed (non-i.i.d.) data settings, we analytically characterize the convergence bound of HIST for non-convex loss functions, and explore the performance-resource trade-off. We also provide guidelines on designing the key system parameters of HFL.
- In simulations, we evaluate the effectiveness of the proposed algorithm by training a neural network in two different data distribution setups for hierarchical networks. We show that our proposed HIST achieves significant resource savings compared with the hierarchical FedAvg.

Related Works: The exploration of submodel training commenced with the pioneering work [5], where the authors introduced the concept of IST for fully connected networks and provided theoretical analysis under centralized settings. Subsequently, submodel training was extended to graph neural networks [6] and ResNets [7]. Due to its effectiveness in communication, computation, and storage, the concept of submodel training was also studied in distributed scenarios [8], where the authors empirically show the effectiveness of submodel training in FL. A series of studies also characterize the convergence behavior of distributed submodel training [9]–[11]. However, the aforementioned works either rely on

restrictive assumptions [9], [10] or narrow the focus to quadratic models [11]. More importantly, existing works focus on cloud-based FL with a single server, and thus do not provide insights into the hierarchical case. To the best of our knowledge, HIST is one of the earliest works to integrate IST with HFL and provide theoretical analysis.

II. System Model and Motivation

We consider a HFL system that consists of a single cloud server, N edge servers indexed by $\{1, 2, \ldots, N\}$, and $\sum_{j=1}^{N} n_j$ clients, where n_j is the number of clients located in the j-th cell. Edge server j is responsible for coordinating the training of n_j clients in cell j. The global server is in charge of model aggregation over N geographically distributed edge servers. Given the loss function $l(\boldsymbol{x}, \xi_i)$ which measures the loss on sample ξ_i with model $\boldsymbol{x} \in \mathbb{R}^d$, the optimization objective of this HFL system can be formulated as

$$\begin{aligned} \min_{\boldsymbol{x}} f(\boldsymbol{x}) &:= \frac{1}{N} \sum_{j=1}^{N} f_j(\boldsymbol{x}) && \text{Global loss} \\ f_j(\boldsymbol{x}) &:= \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} F_i(\boldsymbol{x}) && \text{Cell loss} \\ F_i(\boldsymbol{x}) &:= \mathbb{E}_{\xi_i \sim \mathcal{D}_i} \left[l(\boldsymbol{x}, \xi_i) \right] && \text{Client loss} \end{aligned} \tag{1}$$

where $f : \mathbb{R}^d \leftarrow \mathbb{R}$, $f_j : \mathbb{R}^d \leftarrow \mathbb{R}$, and $F_i : \mathbb{R}^d \leftarrow \mathbb{R}$ represent the global, cell, client losses, respectively. $\mathcal{C}_j$ denotes the client set of cell j, and $\mathcal{D}_i$ denotes the local data distribution of client i. In this work, we mainly consider the non-i.i.d. scenario that the data distribution is heterogeneous among different clients, i.e., $\mathcal{D}_i \neq \mathcal{D}_j$, $\forall i \neq j$.

In conventional HFL, all clients in the system are required to train the full model. To support such model training, each client needs to be equipped with enough computation, storage, and communication resources. However, it is unaffordable for the resource-constrained mobile and Internet of Things (IoT) devices to handle the training of large-scale models. This motivates us to develop a more efficient training framework for HFL, which will be discussed in the next section.

III. Hierarchical Federated Independent Submodel Training Algorthm

Inspired by IST [5], we develop a hierarchical federated submodel training algorithm termed HIST, by incorporating HFL and submodel partitioning. The overview of HIST is described in Fig. 1 and Algorithm 1. The global cloud server periodically aggregates the models from the edge servers, while each edge server periodically aggregates the models from the clients within the corresponding cell. The key difference with the conventional HFL is that, clients do not need to store, update, and exchange the full model in HIST.

Specifically, in the beginning of t/E-th global round where t represents the iteration number of clients and E

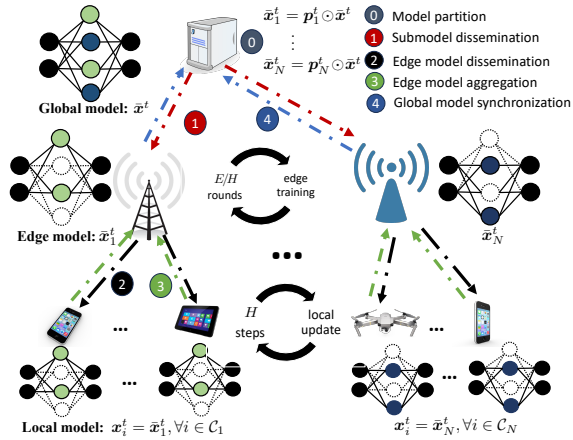


Fig. 1: Overview of proposed hierarchical independent submodel training (HIST). Each cell is responsible for training only a specific partition of the full model, where the submodel partitioning changes over global rounds.

denotes the period of the global aggregation, the cloud server initiates the training process by partitioning the current global model $\bar{\boldsymbol{x}}^t$ into N disjoint submodels:

$$\bar{\boldsymbol{x}}_j^t = \boldsymbol{p}_j^t \odot \bar{\boldsymbol{x}}^t, \forall j \in \{1, 2, \ldots, N\}, \tag{2}$$

where $\bar{\boldsymbol{x}}_j^t$ represents the j-th submodel and $\boldsymbol{p}_j^t$ is a mask that satisfies

$$\boldsymbol{p}_j^t \cap \boldsymbol{p}_{j'}^t = \emptyset, \forall j' \neq j, \text{ and } \sum_{i=1}^{N} \boldsymbol{p}_j^t = \mathbf{1}. \tag{3}$$

These submodels are then distributed to the edge servers, and each edge server j broadcasts submodel $\bar{\boldsymbol{x}}_j^t$ to the clients within its coverage, such that $\boldsymbol{x}_i^t = \bar{\boldsymbol{x}}_j^t, \forall i \in \mathcal{C}_j$. Once the clients receive the most recent model from the server, they initiate training with their local datasets. The essential steps performed by clients, edge servers, and the global server in our proposed algorithm are outlined as follows:

- **Clients:** Clients first compute stochastic gradients with respect to their corresponding submodels, and then update the local models for H steps via the following sparsified stochastic gradient descent:

  $$\boldsymbol{x}_i^{t+1} = \boldsymbol{x}_i^t - \gamma \boldsymbol{p}_j^t \odot \nabla l(\boldsymbol{x}_i^t, \xi_i), \forall i \in \mathcal{C}_j. \tag{4}$$

  Subsequently, clients upload the updated submodels to the edge server for edge model aggregation.
- **Edge Servers:** After H steps of local submodel updates, each edge server aggregates the local models within its coverage as

  $$\bar{\boldsymbol{x}}_j^{t+1} \leftarrow \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{x}_i^{t+1}, \forall j. \tag{5}$$

  Subsequently, edge servers determine whether to upload the aggregated model to the cloud server or broadcast it to the clients. The criterion is whether the current iteration number t of clients is divisible

Algorithm 1: Hierarchical Federated Independent Submodel Training Algorthm

Input: Initial masks $\{\boldsymbol{p}_1^0, \boldsymbol{p}_2^0, \ldots, \boldsymbol{p}_N^0\}$, model $\bar{\boldsymbol{x}}^0$, $\boldsymbol{x}_i^0 = \bar{\boldsymbol{x}}_j^0 = \boldsymbol{p}_j^0 \odot \bar{\boldsymbol{x}}^0, \forall i \in \mathcal{C}_j, \forall j$, learning rate γ,

```
for t ∈ {0, 1, ..., T − 1} do
    for each cell C_j in parallel do
        for each client i ∈ C_j in parallel do
            Update local submodel by (4)
        end
        if H | t + 1 then
            Edge server j aggregates updated
              local submodels by (5)
            if E | t + 1 then
                Upload x̄_j^{t+1} to the cloud server
            else
                Disseminate to clients
                  x_i^{t+1} ← x̄_j^t, ∀i ∈ C_j
            end
        end
    end
    {p_1^{t+1}, p_2^{t+1}, ..., p_N^{t+1}} ← {p_1^t, p_2^t, ..., p_N^t}
    if E | t + 1 then
        Global synchronization: x̄^{t+1} ← Σ_j^N x̄_j^{t+1}
        Generate masks {p_1^{t+1}, p_2^{t+1}, ..., p_N^{t+1}}
          satisfying Σ_{i=1}^N p_j^{t+1} = 1
        Initialize the clients' model for the next
          round of training:
          x_i^{t+1} ← p_j^{t+1} ⊙ x̄^{t+1}, ∀j, ∀i ∈ C_j [1]
    end
end
```

by E. If not, edge servers just broadcast the edge models in (5) to the corresponding clients; otherwise, edge servers upload their edge models to the cloud server.

- **Cloud Server:** If the client's current iteration number t is a multiple of E, the cloud server aggregates the edge models from edge servers according to

$$\bar{\boldsymbol{x}}^{t+1} \leftarrow \sum_{j}^{N} \bar{\boldsymbol{x}}_j^{t+1}. \tag{6}$$

Subsequently, the cloud server partitions the global model $\bar{\boldsymbol{x}}^{t+1}$ by $\boldsymbol{x}_j^{t+1} \leftarrow \boldsymbol{p}_j^{t+1} \odot \bar{\boldsymbol{x}}^{t+1}$ based on the set of new masks $\{\boldsymbol{p}_1^{t+1}, \boldsymbol{p}_2^{t+1}, \ldots, \boldsymbol{p}_N^{t+1}\}$. Finally, $\boldsymbol{x}_j^{t+1}$ is transmitted to edge server j, $\forall j \in \{1, 2, \ldots, N\}$.

Additionally, it is worth emphasizing that $\bar{\boldsymbol{x}}^t$ is defined on $t \in \{mE \mid m \text{ is any iterger}\}$ and $\bar{\boldsymbol{x}}_j^t$ on $t \in \{mH \mid m \text{ is any iterger}\}$.

With the proposed algorithm, clients and edge servers are not required to store or manipulate the full model parameter. Hence, the computation, communication, and storage burdens for clients and edge servers are going to be released.

IV. Convergence Analysis

In this section, we provide convergence analysis for the proposed HIST algorithm. Although the proposed HIST algorithm shares a similar training process with the hierarchical FedAvg, we stress that the convergence proof for the latter one cannot be directly extended to our case that adopts *submodel partitioning*, due to the effect of the masks. Specifically, when comparing $\boldsymbol{p}_j^t \odot \nabla F_i(\boldsymbol{p}_j^t \odot \boldsymbol{x})$ and $\nabla F_i(\boldsymbol{x})$, the mask $\boldsymbol{p}_j^t$ compresses not only the gradient but also the model, while many existing works only investigate compressing the gradient. Theoretical analysis on the methods of compressing the model [12] is quite limited. Even in the single-cell scenario, existing proofs of IST [9], [10] rely on some stronger assumptions. Finally, the hierarchical architecture with both multiple steps of client update and multiple steps of edge training we consider makes the analysis further complicated.

A. Assumptions

For analysis, we focus on a general non-convex loss function and consider a non-i.i.d data setting. Our theoretical analysis relies on the following assumptions.

Assumption 1. *F_i is differentiable and L-smooth, i.e., there exists a positive constant L such that*

$$\begin{aligned} &\|\nabla F_i(\boldsymbol{x}) - \nabla F_i(\boldsymbol{y})\|^2 \leq L\|\boldsymbol{y} - \boldsymbol{x}\|, \forall i, \\ &F_i(\boldsymbol{y}) \leq F_i(\boldsymbol{x}) + \langle \nabla F_i(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x} \rangle + \frac{L}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2. \end{aligned} \tag{7}$$

With Assumption 1, one can also claim that functions f_j, $\forall j$ and f are L-smooth [13].

Assumption 2. *The stochastic gradient $\nabla l(\boldsymbol{x}, \xi_i)$ is an unbiased estimate of the true gradient, i.e., $\mathbb{E}_{\xi_i \sim \mathcal{D}_i}[\nabla l(\boldsymbol{x}, \xi_i)] = \nabla F_i(\boldsymbol{x})$.*

Assumption 3. *There exists a constant σ^2 such that*

$$\mathbb{E}_{\boldsymbol{\xi}_i \sim \mathcal{D}_i}\|\nabla l(\boldsymbol{x}, \xi_i) - \nabla F_i(\boldsymbol{x})\|^2 \leq \sigma^2. \tag{8}$$

Assumption 4. *The gradient dissimilarity between the global loss and each edge loss f_j can be bounded by a constant δ_1^2, i.e.,*

$$\|\nabla f_j(\boldsymbol{x}) - \nabla f(\boldsymbol{x})\|^2 \leq \delta_1^2, \forall \boldsymbol{x}, \forall j. \tag{9}$$

Assumption 5. *The gradient dissimilarity between the edge loss f_j and each client loss $F_i(\boldsymbol{x})$ can be bounded by a constant δ_2^2, i.e.,*

$$\|\nabla F_i(\boldsymbol{x}) - \nabla f_j(\boldsymbol{x})\|^2 \leq \delta_2^2, \forall \boldsymbol{x}, \forall j, \forall i \in \mathcal{C}_j. \tag{10}$$

Remark 1. *Assumptions 1-3 have been adopted in the literature on stochastic non-convex and smooth settings [13]. Assumptions 4 and 5 characterize the extent of inter-cell and intra-cell data heterogeneity, respectively, which are common within the HFL literature [4].*

B. Theoretical Results

We establish the convergence properties of the proposed algorithm by characterizing the evolution of $\|\nabla f(\bar{\boldsymbol{x}}^t)\|^2$, to see how fast the model converges to the stationary point of the general non-convex loss function. When implementing HIST in practice, it is worth noting that $\bar{\boldsymbol{x}}^t$ will not be computed unless $E \mid t$ holds, as the global synchronization occurs every E iterations. The terms $\{\bar{\boldsymbol{x}}^t \mid t = 1, 2, \ldots, T\}$ we use for analysis serve as virtual sequences commonly employed to monitor the convergence of distributed algorithms with multiple local updates. Now we state the following main theorem. Due to space limitation, all proofs are available in our technical report [add citation with a link].

Theorem 1. *Suppose that Assumptions 1-5 hold and the step size satisfies*

$$\gamma \leq \min\left\{\frac{1}{HL\sqrt{24}}, \frac{1}{E\sqrt{324(N-1)+72(N^2+1)L^2}}, \frac{1}{2(N+1)E}, \frac{L}{2(N+1)E}, \frac{1}{EL\sqrt{80(N^2+1)}}\right\}. \tag{11}$$

Then, HIST achieves the following convergence behavior for non-convex loss functions:

$$\begin{aligned}
&\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 \\
\leq& 4\frac{f(\bar{\boldsymbol{x}}_0)-\mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma T} + 2\gamma L\sigma^2\tilde{N} + 12\gamma\sigma^2\tilde{N} \\
&+ 48\gamma^2 NH\sigma^2 + 144\gamma^2 NH^2\delta_2^2 + 6N\delta_1^2 + 81\delta_1^2 \\
&+ 8(N-1)E\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\|\bar{\boldsymbol{x}}_{m'E}\|^2,
\end{aligned} \tag{12}$$

where $\tilde{N} = \sum_{j=1}^{N}\frac{1}{n_j}$, and $\{\bar{\boldsymbol{x}}_{m'E} | m' = 1, 2, \ldots, T/E\}$ are the models generated by our HIST algorithm.

Theorem 1 presents the optimality gap for the time-averaged squared gradient norm. The first term in this upper bound exhibits the effect of the initial optimality gap on convergence performance. The second, third, and fourth terms reveal the impact of the variance of stochastic gradients on convergence, which can be reduced by increasing the mini-batch size when computing stochastic gradients. The fifth to seventh terms show that the non-i.i.d. characteristics across cells and within the cell affect convergence. The last term demonstrates that the norms of synchronized global models also influence the optimality gap. In addition, the step size γ is a configurable parameter that impacts the first five terms of this upper bound. Plugging an appropriate step size into Theorem 1 gives rise to the following corollary.

Corollary 1. *Suppose that Assumptions 1-5 hold and let the step size $\gamma = (T\tilde{N})^{-\frac{1}{2}}$ in which T is large enough to satisfy both (11) and*

$$\gamma \leq (NH^2)^{-1}. \tag{13}$$

Then, the models $\{\bar{\boldsymbol{x}}^t \mid t = 1, 2, \ldots, T\}$ generated by hierarchical federated independent submodel training method satisfies

$$\begin{aligned}
\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 &\leq \mathcal{O}\left(\tilde{N}^{\frac{1}{2}}T^{-\frac{1}{2}}\right) + 6N\delta_1^2 + 81\delta_1^2 \\
&+ 8(N-1)E\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\|\bar{\boldsymbol{x}}_{m'E}\|^2,
\end{aligned} \tag{14}$$

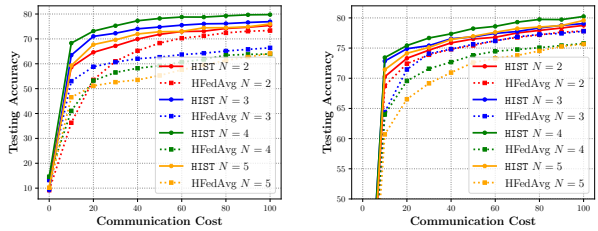
where $\tilde{N} = \sum_{j=1}^{N}\frac{1}{n_j}$.

Remark 2. *In Corollary 1, we keep $\tilde{N}$ in the convergence rate expression as the relationship between n_j and N could be arbitrary. If the number of clients in each cell n_j is in the same order or bigger than the number of cells N, $\tilde{N}$ will be no more than $\mathcal{O}(1)$ and the convergence rate can be expressed as $\mathcal{O}\left(T^{-\frac{1}{2}}\right)$.*

Non-diminishing bound: With the step size chosen in Corollary 1, the first five terms in (12) will diminish to zero as long as the number of total iterations, i.e., T, is large enough. The diminishing rate of the five terms can be expressed as $\mathcal{O}\left(\tilde{N}^{\frac{1}{2}}T^{-\frac{1}{2}}\right)$. The rest three terms are non-diminishing parts that arise due to submodel training. One can claim that hierarchical federated independent submodel training can converge to the neighborhood of a stationary point of the non-convex loss function under appropriate conditions. A similar phenomenon has also been reported in the single-cell case [5], [9], [11]. The bound enables us to explore the performance-resource trade-off, where more detailed discussions will be provided in the next paragraph.

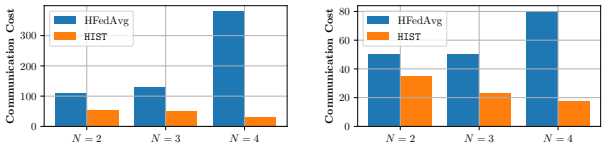
The choice of N: As N increases, i.e., as the overall clients in the system are divided into more cells during training, the size of the submodels get smaller, providing a more lightweight model to the edge servers and clients. As a result, the training costs including computation, communication, and storage will get reduced at each iteration. However, as observed in Corollary 1, a large N causes the sequence to deviate further from the stationary point. Overall, there is a trade-off between the convergence performance and computation, communication, and storage costs.

The optimal values of H and E: The choices of H and E impact the communication frequency. As H increases, the aggregation frequency at the edge servers will become smaller, reducing the communication load between clients and the edge server. On the other hand, a large E induces fewer global synchronizations, which releases the communication burden between edge servers and the cloud server. However, these values cannot be



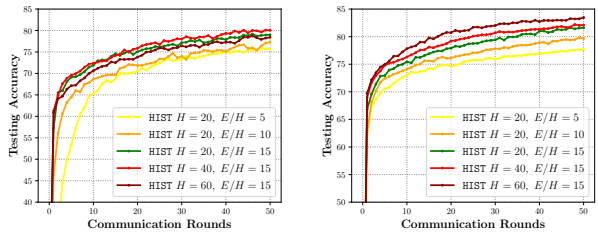
(a) Non-i.i.d. data (b) Mixed data

Fig. 2: The impact of the number of cells, i.e., N, on the convergence performance



(a) Non-i.i.d. data (b) Mixed data

Fig. 3: Communication cost for achieving a testing accuracy of 75%



(a) Non-i.i.d. data (b) Mixed data

Fig. 4: The impact of the number of local updates, i.e., H and E, on the convergence performance

infinitely large. The maximum value of H and E can be derived from the condition of the step size γ. Specifically, to make the step size $\gamma = (T\tilde{N})^{-\frac{1}{2}}$ in Corollary 1 satisfy inequalities (11) and (13), H and E can be set as $\mathcal{O}\left((\tilde{N}T)^{\frac{1}{4}}N^{-\frac{1}{2}}\right)$ and $\mathcal{O}\left((\tilde{N}T)^{\frac{1}{2}}N^{-1}\right)$, respectively.

C. Communication Complexity Analysis

Let $|X|$ denote the full model size. Each client sends its local model parameter to the corresponding edge server every H iterations, where H denotes the number of local updates. Assume mask size, the number of non-zero entries of $\boldsymbol{p}_j$, is uniform among N cells. Every H iterations, the total communication load of all the clients within cell j is $\frac{n_j|X|}{N}$, which corresponds to the communication complexity of the edge server j. The average per-iteration communication load of each client is $\frac{|X|}{NH}$. For the cloud server, the communication complexity of every E iterations is $|X|$. One can thus claim that the per-iteration complexity of the cloud server is $\frac{|X|}{E}$.

V. Simulations

In this section, we conduct experiments to evaluate the performance of the proposed HIST algorithm.

A. Simulation Settings

We consider an image classification task on Fashion-MNIST using a two-layer fully connected neural network. In this model, we configure the input layer to have 784 neurons, corresponding to the size of the input image, and the output layer to have 10 neurons, which matches the number of classes. Additionally, we employ a hidden layer with 300 neurons. The cloud server partitions these hidden neurons to construct different submodels. Let the submodels share the same size with each other, which can be achieved by uniformly and randomly partitioning the hidden neurons.

Set a total of 60 clients, evenly distributed across N cells, where N takes value from $\{2, 3, 4, 5\}$. The following experiments will cover two scenarios: fully non-i.i.d. and a combination of non-i.i.d. within intra-cell levels alongside i.i.d. within inter-cell levels. For the former case, the client's dataset construction follows the approach outlined in [14]. The process starts by sorting the training samples according to their respective labels and then dividing the training dataset into 120 shards, each shard containing 500 samples. Subsequently, 2 shards are allocated to each client, ensuring that each client has a dataset containing 1000 samples. Each client is thus assigned at most four unique image labels. For the latter, we first uniformly and randomly divide the entire training data set into N parts, and then distribute these N data sets to clients within N cells in a non-i.i.d. manner, which is similar to the former case.

B. Experiment Results

In Fig. 2, we present a comparison between the proposed HIST algorithm and the traditional hierarchical FedAvg, denoted as HFedAvg in our figures. We compare their performance in terms of testing accuracy under different numbers of cells, $N \in \{2, 3, 4, 5\}$. The x-axis here represents the communication load which quantifies the volume of parameters transmitted by each client, where the unit is set to the load of a full-model transmission. For each client, the communication cost per global round is equal to $\frac{1}{N}\frac{E}{H}$ times the load of a full-model transmission. Here, E/H represents the number of edge aggregations per global round. We set the values of H and E to 40 and 200, respectively. As shown in Figs. 2a and 2b, the proposed HIST algorithm outperforms hierarchical FedAvg in terms of testing accuracy at the same levels of communication consumption for both non-i.i.d. and mixed data settings. Additionally, as N increases, the non-i.i.d. extent of data among clients becomes more pronounced, leading to a performance degeneration for hierarchical FedAvg. In contrast, the proposed HIST achieves a higher testing accuracy when N increases from $N = 2$ to $N = 3$ to $N = 4$. This is due to

the fact that, for HIST, the per-round communication cost per client decreases inversely with the number of cells. However, when we increase the number of cells to $N = 5$, HIST also suffers a performance degeneration. This can be attributed to the submodel getting too small to effectively handle this task, highlighting the existence of a tradeoff between the number of cells and testing accuracy. It is in line with our theoretical results.

Fig. 3 compares the communication cost of HIST and hierarchical FedAvg for achieving a desired accuracy under both non-i.i.d. and mixed data distribution. This experiment was carried out in the setting of $H = 40$ and $E = 200$. The desired testing accuracy is set to be 75%. The Y-axis measures the size of parameters transmitted by each client during the training process, where the unit is set to the load of a full-model transmission. It is observed that the HIST algorithm takes less communication cost to achieve the preset accuracy, which demonstrates the efficiency of the proposed HIST algorithm in hierarchical FedAvg. In addition, as the number of cells increases from $N = 2$ to $N = 3$ to $N = 4$, the communication cost shows a decreasing trend, which forms a sharp comparison with hierarchical FedAvg. This further demonstrates the advantage of HIST.

The impact of the periods of the edge aggregation, i.e., H, and of the global synchronization, i.e., E/H on the convergence behavior is demonstrated in Fig. 4. The x-axis represents the global communication round, also known as the number of global model synchronization. A higher value of H corresponds to a lower communication frequency between clients and edge servers, while a higher E/H ratio indicates a lower communication frequency between edge servers and the cloud server. We consider the 3-cell case (i.e., $N = 3$) where 60 users are uniformly distributed in 3 disjoint cells. As the value of E/H increases from 5 to 10 to 15, HIST attain a better convergence performance, which is witnessed by both Figs. 4a and 4b. This is because a large value of E/H gives rise to a lower communication load for each round. When H increases from 20 to 40 to 60, Fig. 4b shows that the convergence speed HIST first enjoys an acceleration and then a degradation. The latter is induced by data heterogeneity. This phenomenon also fits well with our theory, where we show that there is an upper bound for the number of local updates. Fig. 4b shows that HIST has a better convergence performance as H increases from 40 to 60. This is because mixed data distribution exhibits lower data heterogeneity, which allows for a larger number of local updates.

VI. Conclusion

In this paper, we develop a communication-, computation-, and storage-efficient hierarchical federated training algorithm, named HIST, by incorporating independent model training and local training. We investigate its convergence behavior with arbitrary submodel partition under non-convex loss functions and non-i.i.d. data settings, characterizing the impact of non-i.i.d. extent, the number of periods of edge and global aggregation, and the number of cells on the convergence performance. We show that the HIST algorithm converges with rate $\mathcal{O}\left(\tilde{N}^{\frac{1}{2}}T^{-\frac{1}{2}}\right)$ to a neighborhood of a stationary point of the global loss function. Simulation results illustrate the performance gains with respect to model accuracy and the communication efficiency achieved by the proposed HIST compared to the standard hierarchical FedAvg.

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APPENDIX

Before the proof, we introduce some notations. $\bar{\boldsymbol{x}}_j^t = \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{x}_i^t$ denotes the virtual aggregation of client parameters within cell j. $\bar{\boldsymbol{x}}^t := \sum \boldsymbol{p}_j^t \odot \bar{\boldsymbol{x}}_j^t$ denotes the virtual global aggregation.

We analyze the iteration of the virtual sequence $\bar{\boldsymbol{x}}_t$. Based on the L-smooth assumption, i.e., Assumption 1, we have

$$\mathbb{E}[f(\bar{\boldsymbol{x}}_{t+1})] \leq f(\bar{\boldsymbol{x}}_t) \underbrace{-\mathbb{E}\left\langle \nabla f(\bar{\boldsymbol{x}}_t), \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\gamma \boldsymbol{g}_i^t \right\rangle}_{T_1} + \underbrace{\frac{L}{2}\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\gamma \boldsymbol{g}_i^t \right\|^2}_{T_2}. \tag{15}$$

For T_2, we have

$$\begin{aligned} T_2 =& \mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\gamma \boldsymbol{g}_i^t \right\|^2 \\ =& \gamma^2\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j} \boldsymbol{g}_i^t - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 + \gamma^2\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \end{aligned} \tag{16}$$

As there is no overlapping between any two different masks in the same round, we have

$$\begin{aligned} T_2 =& \gamma^2 \sum_{j=1}^N \mathbb{E}\left\| \frac{1}{n_j}\sum_{i\in\mathcal{C}_j} \boldsymbol{g}_i^t - \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 + \gamma^2\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \\ =& \gamma^2 \sum_{j=1}^N \frac{1}{n_j^2}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\| \boldsymbol{g}_i^t - \nabla F_i(\boldsymbol{x}_i^t) \right\|^2 + \gamma^2\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \\ \leq& \gamma^2\sigma^2\left(\sum_{j=1}^N \frac{1}{n_j}\right) + \gamma^2\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2, \end{aligned} \tag{17}$$

where the inequality comes from Assumption 3.

For T_1, we have

$$\begin{aligned} T_1 =& -\gamma\mathbb{E}\left\langle \nabla f(\bar{\boldsymbol{x}}_t), \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j} \boldsymbol{g}_i^t \right\rangle \\ =& -\gamma\mathbb{E}\left\langle \nabla f(\bar{\boldsymbol{x}}_t), \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right\rangle \\ =& \frac{\gamma}{2}\left\{ \mathbb{E}\left\| \nabla f(\bar{\boldsymbol{x}}_t) - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 - \|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 - \mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \right\}. \end{aligned} \tag{18}$$

Plugging T_1 and T_2 into (15) gives the following inequality.

$$\begin{aligned} \mathbb{E}[f(\bar{\boldsymbol{x}}_{t+1})] \leq& f(\bar{\boldsymbol{x}}_t) - \frac{\gamma}{2}\|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 - \left(\frac{\gamma}{2} - \frac{L\gamma^2}{2}\right)\mathbb{E}\left\| \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \\ &+ \frac{\gamma}{2}\mathbb{E}\left\| \nabla f(\bar{\boldsymbol{x}}_t) - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2 + \frac{\gamma^2 L\sigma^2}{2}\left(\sum_{j=1}^N \frac{1}{n_j}\right) \end{aligned} \tag{19}$$

As $\gamma \leq \frac{1}{L}$, we have

$$\mathbb{E}[f(\bar{\boldsymbol{x}}_{t+1})] \leq f(\bar{\boldsymbol{x}}_t) - \frac{\gamma}{2}\|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 + \frac{\gamma^2 L\sigma^2}{2}\left(\sum_{j=1}^N \frac{1}{n_j}\right) + \frac{\gamma}{2}\underbrace{\mathbb{E}\left\| \nabla f(\bar{\boldsymbol{x}}_t) - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t) \right\|^2}_{T_3} \tag{20}$$

We next bound T_3 as follows.

$$
\begin{aligned}
T_3 =& \sum_{j=1}^{N} \mathbb{E} \left\| \boldsymbol{p}_j^t \odot \left(\nabla f(\bar{\boldsymbol{x}}_t) \mp \nabla f_j(\bar{\boldsymbol{x}}_t) \mp \nabla f_j(\bar{\boldsymbol{x}}_t^j) - \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
\leq& 3 \sum_{j=1}^{N} \mathbb{E} \left\| \boldsymbol{p}_j^t \odot \nabla f(\bar{\boldsymbol{x}}_t) - \boldsymbol{p}_j^t \odot \nabla f_j(\bar{\boldsymbol{x}}_t) \right\|^2 + 3 \sum_{j=1}^{N} \mathbb{E} \left\| \boldsymbol{p}_j^t \odot \nabla f_j(\bar{\boldsymbol{x}}_t) - \boldsymbol{p}_j^t \odot \nabla f_j(\bar{\boldsymbol{x}}_t^j) \right\|^2 \\
&+ 3 \sum_{j=1}^{N} \mathbb{E} \left\| \boldsymbol{p}_j^t \odot \nabla f_j(\bar{\boldsymbol{x}}_t^j) - \boldsymbol{p}_j^t \odot \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \\
\leq& 3N\delta_1^2 + 3 \sum_{j=1}^{N} \mathbb{E} \left\| \bar{\boldsymbol{x}}_t - \bar{\boldsymbol{x}}_t^j \right\|^2 + 3 \sum_{j=1}^{N} \mathbb{E} \left\| \boldsymbol{p}_j^t \odot \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\bar{\boldsymbol{x}}_t^j) - \boldsymbol{p}_j^t \odot \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \\
\leq& 3N\delta_1^2 + 3 \sum_{j=1}^{N} \mathbb{E} \left\| \bar{\boldsymbol{x}}_t - \bar{\boldsymbol{x}}_t^j \right\|^2 + 3 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^t \odot \nabla F_i(\bar{\boldsymbol{x}}_t^j) - \boldsymbol{p}_j^t \odot \nabla F_i(\boldsymbol{x}_i^t) \right\|^2 \\
\leq& 3N\delta_1^2 + \underbrace{3 \sum_{j=1}^{N} \mathbb{E} \left\| \bar{\boldsymbol{x}}_t - \bar{\boldsymbol{x}}_t^j \right\|^2}_{T_6} + \underbrace{3 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \bar{\boldsymbol{x}}_t^j - \boldsymbol{x}_i^t \right\|^2}_{T_7}
\end{aligned}
\tag{21}
$$

where the first equality follows from $\sum_{j=1}^{N} \boldsymbol{p}_j^t = \mathbf{1}$, the first inequality comes from Cauchy-Swarchz inequality, the second and the fourth inequalities come from Assumption 1, and the third inequality follows Jensen's inequality. Therefore,

$$
\mathbb{E}[f(\bar{\boldsymbol{x}}_{t+1})] \leq f(\bar{\boldsymbol{x}}_t) - \frac{\gamma}{2} \left\| \nabla f(\bar{\boldsymbol{x}}_t) \right\|^2 + \frac{\gamma^2 L \sigma^2}{2} \left(\sum_{j=1}^{N} \frac{1}{n_j} \right) + \frac{3\gamma}{2} N\delta_1^2 + \frac{3\gamma}{2} T_6 + \frac{3\gamma}{2} T_7 \tag{22}
$$

Reorganizing the above terms, we have

$$
\left\| \nabla f(\bar{\boldsymbol{x}}_t) \right\|^2 \leq 2 \frac{f(\bar{\boldsymbol{x}}_t) - \mathbb{E}[f(\bar{\boldsymbol{x}}_{t+1})]}{\gamma} + \gamma L \sigma^2 \left(\sum_{j=1}^{N} \frac{1}{n_j} \right) + 3N\delta_1^2 + 3T_6 + 3T_7. \tag{23}
$$

Taking expectation and telescoping the above inequalities from $t = 0$ to $T - 1$, we have

$$
\begin{aligned}
\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\| \nabla f(\bar{\boldsymbol{x}}_t) \right\|^2 \leq& 2 \frac{f(\bar{\boldsymbol{x}}_0) - \mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma} + \gamma L \sigma^2 \left(\sum_{j=1}^{N} \frac{1}{n_j} \right) + 3N\delta_1^2 \\
&+ 3 \frac{1}{T} \sum_{t=0}^{T-1} T_6 + 3 \frac{1}{T} \sum_{t=0}^{T-1} T_7.
\end{aligned}
\tag{24}
$$

Next, we need to bound T_6 and T_7. Without the loss generality, we consider $mE < t < (m+1)E$. T_6 can be

bounded as follows.

$$
\begin{aligned}
T_6 =& \sum_{j=1}^{N} \mathbb{E} \left\| \bar{\boldsymbol{x}}_t - \bar{\boldsymbol{x}}_t^j \right\|^2 \\
=& \sum_{j=1}^{N} \mathbb{E} \left\| \bar{\boldsymbol{x}}_{mE} - \gamma \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot g_i^\tau - \left(\bar{\boldsymbol{x}}_{mE}^j - \gamma \sum_{\tau=mE}^{t} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot g_i^\tau \right) \right\|^2 \\
\leq& 2 \sum_{j=1}^{N} \left\| \bar{\boldsymbol{x}}_{mE} - \bar{\boldsymbol{x}}_{mE}^j \right\|^2 + 2\gamma^2 \sum_{j=1}^{N} \mathbb{E} \left\| \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot g_i^\tau - \sum_{\tau=mE}^{t} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot g_i^\tau \right\|^2 \\
=& 2\gamma^2 \mathbb{E} \sum_{j=1}^{N} \left\| \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot (g_i^\tau \mp \nabla F_i(\boldsymbol{x}_i^\tau)) - \sum_{\tau=mE}^{t} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot (g_i^\tau \mp \nabla F_i(\boldsymbol{x}_i^\tau)) \right\|^2 \\
& 2 \sum_{j=1}^{N} \left\{ \left\| \boldsymbol{p}_j^{mE} \odot \bar{\boldsymbol{x}}_{mE} \right\|^2 + \left\| \bar{\boldsymbol{x}}_{mE} \right\|^2 - 2 \left\langle \boldsymbol{p}_j^{mE} \odot \bar{\boldsymbol{x}}_{mE}, \bar{\boldsymbol{x}}_{mE} \right\rangle \right\} \\
\leq& 6\gamma^2 \sum_{j=1}^{N} \mathbb{E} \left\| \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot (g_i^\tau - \nabla F_i(\boldsymbol{x}_i^\tau)) \right\|^2 + 6\gamma^2 \sum_{j=1}^{N} \mathbb{E} \left\| \sum_{\tau=mE}^{t} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot (g_i^\tau - \nabla F_i(\boldsymbol{x}_i^\tau)) \right\|^2 \\
& + 6\gamma^2 \sum_{j=1}^{N} \mathbb{E} \left\| \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot \nabla F_i(\boldsymbol{x}_i^\tau) - \sum_{\tau=mE}^{t} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot \nabla F_i(\boldsymbol{x}_i^\tau) \right\|^2 + 2(N-1) \left\| \bar{\boldsymbol{x}}_{mE} \right\|^2 \\
=& 6\gamma^2 N \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j^2} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot (g_i^\tau - \nabla F_i(\boldsymbol{x}_i^\tau)) \right\|^2 + 6\gamma^2 \sum_{j=1}^{N} \sum_{\tau=mE}^{t} \frac{1}{n_j^2} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot (g_i^\tau - \nabla F_i(\boldsymbol{x}_i^\tau)) \right\|^2 \\
& + 6\gamma^2 \sum_{j=1}^{N} \mathbb{E} \left\| \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot \nabla F_i(\boldsymbol{x}_i^\tau) - \sum_{\tau=mE}^{t} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot \nabla F_i(\boldsymbol{x}_i^\tau) \right\|^2 + 2(N-1) \left\| \bar{\boldsymbol{x}}_{mE} \right\|^2 \\
\leq& 6\gamma^2 E \sum_{\tau=mE}^{t} \sum_{j=1}^{N} \underbrace{\mathbb{E} \left\| \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot \nabla F_i(\boldsymbol{x}_i^\tau) - \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{p}_j^\tau \odot \nabla F_i(\boldsymbol{x}_i^\tau) \right\|^2}_{T_8} \\
& + 6\gamma^2 (N+1) E \sigma^2 \sum_{j=1}^{N} \frac{1}{n_j} + 2(N-1) \left\| \bar{\boldsymbol{x}}_{mE} \right\|^2,
\end{aligned}
\tag{25}
$$

where the first and second inequalities come from Cauchy-Swartz inequality and the third inequality follows Assumption 3.

For T_8, we have

$$
\begin{aligned}
&\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\boldsymbol{w}_i^{\tau}\right)\mp\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\boldsymbol{w}_i^{\tau}\right)\mp\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\right)\right\|^2\\
\leq&3\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\boldsymbol{w}_i^{\tau}\right)-\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\right)\right\|^2+3\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\right\|^2\\
&+3\mathbb{E}\left\|\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\boldsymbol{w}_i^{\tau}\right)-\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\right)\right\|^2\\
\leq&3NL^2\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^{\tau}-\bar{\boldsymbol{w}}_{\tau}^{j}\right\|^2+3\underbrace{\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\right\|^2}_{T_9}\\
&+3L^2\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^{\tau}-\bar{\boldsymbol{w}}_{\tau}^{j}\right\|^2,
\end{aligned}
\tag{26}
$$

where the first inequality comes from Cauchy-Swartz inequality and the second inequality comes from Jensen's inequality and L-smoothness of F_i. For T_9, we have

$$
\begin{aligned}
T_9=&\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\mp\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)\mp\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)\right\|^2\\
\leq&3\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)-\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)\right\|^2+3\mathbb{E}\left\|\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\left(\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)-\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)\right\|^2\\
&+3\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
\leq&3N\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)-\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2+3\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}^{j}\right)-\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
&+3\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
\leq&3L^2N\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{\tau}^{j}-\bar{\boldsymbol{w}}_{\tau}\right\|^2+3L^2\mathbb{E}\left\|\bar{\boldsymbol{w}}_{\tau}^{j}-\bar{\boldsymbol{w}}_{\tau}\right\|^2\\
&+3\underbrace{\mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2}_{T_{10}}.
\end{aligned}
$$

For T_{10}, we have

$$
\begin{aligned}
T_{10} =& \mathbb{E}\left\|\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{p}_j^{\tau}\odot\nabla F_i\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
=& \mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^{\tau}\odot\nabla f_j\left(\bar{\boldsymbol{w}}_{\tau}\right)-\boldsymbol{p}_j^{\tau}\odot\nabla f_j\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
=& \mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^{\tau}\odot\left(f_j\left(\bar{\boldsymbol{w}}_{\tau}\right)\mp f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)-\boldsymbol{p}_j^{\tau}\odot\left(\nabla f_j\left(\bar{\boldsymbol{w}}_{\tau}\right)\mp\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)\right\|^2\\
\leq& \mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^{\tau}\odot\left(\nabla f_j\left(\bar{\boldsymbol{w}}_{\tau}\right)-\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)\right\|^2+3\mathbb{E}\left\|\boldsymbol{p}_j^{\tau}\odot\left(\nabla f_j\left(\bar{\boldsymbol{w}}_{\tau}\right)-\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right)\right\|^2\\
&+3\mathbb{E}\left\|\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)-\sum_{j=1}^{N}\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
\leq& 3(N+1)\delta_1^2+3\mathbb{E}\left\|\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)-\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2
\end{aligned}
$$

Combining T_6, T_8, T_9, and T_{10}, we can establish an upper for T_6 as follows.

$$
\begin{aligned}
T_6 \leq& 18\gamma^2 EN^2L^2\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^{\tau}-\bar{\boldsymbol{w}}_{\tau}^{j}\right\|^2+18\gamma^2L^2E\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^{\tau}-\bar{\boldsymbol{w}}_{\tau}^{j}\right\|^2\\
&+72\gamma^2EL^2N^2\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{\tau}^{j}-\bar{\boldsymbol{w}}_{\tau}\right\|^2+72\gamma^2EL^2\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{\tau}^{j}-\bar{\boldsymbol{w}}_{\tau}\right\|^2\\
&+162\gamma^2N(N+1)E^2\delta_1^2+162\gamma^2E\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\mathbb{E}\left\|\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)-\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
&+6\gamma^2(N+1)E\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+2(N-1)\left\|\bar{\boldsymbol{w}}_{mE}\right\|^2
\end{aligned}
$$

Besides, we have

$$
\begin{aligned}
\sum_{j=1}^{N}\mathbb{E}\left\|\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)-\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2 &= \sum_{j=1}^{N}\mathbb{E}\left\{\left\|\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2-2\left\langle\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right),\boldsymbol{p}_j^{\tau}\odot\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\rangle+\left\|\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\right\}\\
&= (N-1)\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
T_6 \leq& 18\gamma^2E\left(N^2+1\right)L^2\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^{\tau}-\bar{\boldsymbol{w}}_{\tau}^{j}\right\|^2+72\gamma^2E\left(N^2+1\right)L^2\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{\tau}^{j}-\bar{\boldsymbol{w}}_{\tau}\right\|^2\\
&+162\gamma^2N(N+1)E^2\delta_1^2+162\gamma^2E(N-1)\sum_{\tau=mE}^{t}\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_{\tau}\right)\right\|^2\\
&+6\gamma^2(N+1)E\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+2(N-1)\left\|\bar{\boldsymbol{w}}_{mE}\right\|^2
\end{aligned}
$$

Hence,

$$
\begin{aligned}
&\frac{1}{T}\sum_{t=0}^{T-1} T_6 \leq 18\gamma^2 E\left(N^2+1\right)L^2\frac{1}{T}\sum_{t=0}^{T-1}\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^\tau-\bar{\boldsymbol{w}}_\tau^j\right\|^2 + 72\gamma^2 E\left(N^2+1\right)L^2\frac{1}{T}\sum_{t=0}^{T-1}\sum_{\tau=mE}^{t}\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{w}}_\tau^j-\bar{\boldsymbol{w}}_\tau\right\|^2\\
&+162\gamma^2 N(N+1)E^2\delta_1^2+162\gamma^2 E(N-1)\frac{1}{T}\sum_{t=0}^{T-1}\sum_{\tau=mE}^{t}\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_\tau\right)\right\|^2\\
&+6\gamma^2(N+1)E\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+2(N-1)\frac{E}{T}\sum_{m'=0}^{T/E}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2\\
&\leq 18\gamma^2E^2\left(N^2+1\right)L^2\frac{1}{T}\sum_{t=0}^{T-1}\underbrace{\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^\tau-\bar{\boldsymbol{w}}_\tau^j\right\|^2}_{T_7}+72\gamma^2E^2\left(N^2+1\right)L^2\frac{1}{T}\sum_{t=0}^{T-1}\underbrace{\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{w}}_\tau^j-\bar{\boldsymbol{w}}_\tau\right\|^2}_{T_6}\\
&+162\gamma^2N(N+1)E^2\delta_1^2+162\gamma^2E^2(N-1)\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_\tau\right)\right\|^2\\
&+6\gamma^2(N+1)E\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+2(N-1)\frac{E}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2\\
&\left(1-72\gamma^2E^2\left(N^2+1\right)L^2\right)\frac{1}{T}\sum_{t=0}^{T-1}T_6\leq 18\gamma^2E^2\left(N^2+1\right)L^2\frac{1}{T}\sum_{t=0}^{T-1}\underbrace{\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^\tau-\bar{\boldsymbol{w}}_\tau^j\right\|^2}_{T_7}\\
&+162\gamma^2N(N+1)E^2\delta_1^2+162\gamma^2E^2(N-1)\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_\tau\right)\right\|^2\\
&+6\gamma^2(N+1)E\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+2(N-1)\frac{E}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2
\end{aligned}
\tag{27}
$$

As $\gamma \leq \frac{1}{EL\sqrt{72(N^2+1)}}$, We thus have,

$$
\begin{aligned}
&\frac{1}{T}\sum_{t=0}^{T-1}T_6\leq\frac{18\gamma^2E^2\left(N^2+1\right)L^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{t=0}^{T-1}\underbrace{\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^\tau-\bar{\boldsymbol{w}}_\tau^j\right\|^2}_{T_7}\\
&+\frac{162\gamma^2N(N+1)E^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\delta_1^2+\frac{162\gamma^2E^2(N-1)}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_\tau\right)\right\|^2\\
&+\frac{6\gamma^2(N+1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+\frac{2(N-1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2
\end{aligned}
\tag{28}
$$

For T_7, we have

$$
\begin{aligned}
T_7 =& \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \bar{\boldsymbol{x}}_t^j - \boldsymbol{x}_i^t \right\|^2 \\
=& \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \bar{\boldsymbol{x}}_{t-1}^j - \gamma \boldsymbol{p}_j^t \odot \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{g}_i^{t-1} - \boldsymbol{x}_i^{t-1} + \gamma \boldsymbol{p}_j^\tau \odot \boldsymbol{g}_i^{t-1} \right\|^2 \\
=& \gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \sum_{\tau=\alpha H}^{t-1} \boldsymbol{p}_j^\tau \odot \boldsymbol{g}_i^\tau - \sum_{\tau=\alpha H}^{t-1} \boldsymbol{p}_j^\tau \odot \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{g}_i^\tau \right\|^2 \\
=& \gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \boldsymbol{g}_i^\tau \mp \sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) \mp \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{g}_i^\tau \right) \right\|^2 \\
\leq& 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \boldsymbol{g}_i^\tau - \sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{g}_i^\tau + \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
&+ 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
=& 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \boldsymbol{g}_i^\tau - \sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
&- 2\gamma^2 \sum_{j=1}^{N} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{g}_i^\tau - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
&+ 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
=& 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \sum_{\tau=\alpha H}^{t-1} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\boldsymbol{g}_i^\tau - \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
&- 2\gamma^2 \sum_{j=1}^{N} \sum_{\tau=\alpha H}^{t-1} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{g}_i^\tau - \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
&+ 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
=& 2\gamma^2 \sum_{j=1}^{N} \left(\frac{1}{n_j} - \frac{1}{n_j^2} \right) \sum_{i \in \mathcal{C}_j} \sum_{\tau=\alpha H}^{t-1} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\boldsymbol{g}_i^\tau - \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
&+ 2\gamma^2 \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2 \\
\leq& 2\gamma^2 H \sigma^2 \sum_{j=1}^{N} \left(1 - \frac{1}{n_j} \right) + 2\gamma^2 \sum_{j=1}^{N} \underbrace{\frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbb{E} \left\| \boldsymbol{p}_j^\tau \odot \left(\sum_{\tau=\alpha H}^{t-1} \nabla F_i(\boldsymbol{x}_i^t) - \sum_{\tau=\alpha H}^{t-1} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \nabla F_i(\boldsymbol{x}_i^t) \right) \right\|^2}_{T_8}
\end{aligned}
\tag{29}
$$

Next, we bound T_8.

$$
\begin{aligned}
&\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{p}_j^\tau\odot\left(\sum_{\tau=\alpha H}^{t-1}\nabla F_i(\boldsymbol{x}_i^t)-\sum_{\tau=\alpha H}^{t-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t)\right)\right\|^2\\
=&\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{p}_j^\tau\odot\left(\sum_{\tau=\alpha H}^{t-1}\nabla F_i(\boldsymbol{x}_i^t)\mp\sum_{\tau=\alpha H}^{t-1}\nabla F_i(\bar{\boldsymbol{x}}_t^j)\mp\sum_{\tau=\alpha H}^{t-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_t^j)-\sum_{\tau=\alpha H}^{t-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t)\right)\right\|^2\\
\leq&\frac{3}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{p}_j^\tau\odot\left(\sum_{\tau=\alpha H}^{t-1}\nabla F_i(\boldsymbol{x}_i^t)-\sum_{\tau=\alpha H}^{t-1}\nabla F_i(\bar{\boldsymbol{x}}_t^j)\right)\right\|^2\\
&+\frac{3}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{p}_j^\tau\odot\left(\sum_{\tau=\alpha H}^{t-1}\nabla F_i(\bar{\boldsymbol{x}}_t^j)-\sum_{\tau=\alpha H}^{t-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_t^j)\right)\right\|^2\\
&+\frac{3}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{p}_j^\tau\odot\left(\sum_{\tau=\alpha H}^{t-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_t^j)-\sum_{\tau=\alpha H}^{t-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_i^t)\right)\right\|^2\\
\leq&\frac{6H}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{\tau=\alpha H}^{t-1}\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_i^t)-\nabla F_i(\bar{\boldsymbol{x}}_t^j)\right\|^2+\frac{3H}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{\tau=\alpha H}^{t-1}\mathbb{E}\left\|\nabla F_i(\bar{\boldsymbol{x}}_t^j)-\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_t^j)\right\|^2\\
\leq&\frac{6HL^2}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{\tau=\alpha H}^{t-1}\mathbb{E}\left\|\boldsymbol{x}_i^t-\bar{\boldsymbol{x}}_t^j\right\|^2+3H^2\delta_2^2
\end{aligned}
\tag{30}
$$

Therefore, we have

$$
\begin{aligned}
T_7=&\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\bar{\boldsymbol{x}}_t^j-\boldsymbol{x}_i^t\right\|^2\\
\leq&2\gamma^2NH\sigma^2+6\gamma^2NH^2\delta_2^2+12\gamma^2HL^2\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{\tau=\alpha H}^{t-1}\mathbb{E}\left\|\boldsymbol{x}_i^t-\bar{\boldsymbol{x}}_t^j\right\|^2.
\end{aligned}
\tag{31}
$$

As a result, we obtain

$$
\begin{aligned}
\frac{1}{T}\sum_{t=1}^{T-1}T_7\leq&2\gamma^2NH\sigma^2+6\gamma^2NH^2\delta_2^2+12\gamma^2HL^2\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left(\frac{1}{T}\sum_{t=1}^{T-1}\sum_{\tau=\alpha H}^{t-1}\mathbb{E}\left\|\boldsymbol{x}_i^t-\bar{\boldsymbol{x}}_t^j\right\|^2\right)\\
\leq&2\gamma^2NH\sigma^2+6\gamma^2NH^2\delta_2^2+12\gamma^2H^2L^2\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left(\frac{1}{T}\sum_{t=1}^{T-1}\mathbb{E}\left\|\boldsymbol{x}_i^t-\bar{\boldsymbol{x}}_t^j\right\|^2\right)\\
=&2\gamma^2NH\sigma^2+6\gamma^2NH^2\delta_2^2+12\gamma^2H^2L^2\left(\frac{1}{T}\sum_{t=1}^{T-1}T_7\right)
\end{aligned}
\tag{32}
$$

As $\gamma\leq\frac{1}{HL\sqrt{24}}$, thus $1-12\gamma^2H^2L^2\geq\frac{1}{2}$. We have

$$
\frac{1}{T}\sum_{t=1}^{T-1}T_7\leq4\gamma^2NH\sigma^2+12\gamma^2NH^2\delta_2^2.
\tag{33}
$$

Recalling that

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}T_6 \leq & \frac{18\gamma^2E^2\left(N^2+1\right)L^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{t=0}^{T-1}\underbrace{\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{w}_i^\tau-\bar{\boldsymbol{w}}_\tau^j\right\|^2}_{T_7} \\ & +\frac{162\gamma^2N(N+1)E^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\delta_1^2+\frac{162\gamma^2E^2(N-1)}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f\left(\bar{\boldsymbol{w}}_\tau\right)\right\|^2 \\ & +\frac{6\gamma^2(N+1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\sigma^2\sum_{j=1}^{N}\frac{1}{n_j}+\frac{2(N-1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2\end{aligned} \tag{34}$$

and

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 \leq & 2\frac{f(\bar{\boldsymbol{x}}_0)-\mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma}+\gamma L\sigma^2\left(\sum_{j=1}^{N}\frac{1}{n_j}\right)+3N\delta_1^2 \\ & +3\frac{1}{T}\sum_{t=0}^{T-1}T_6+3\frac{1}{T}\sum_{t=0}^{T-1}T_7\end{aligned} \tag{35}$$

We have

$$\begin{aligned}&\left(1-\frac{162\gamma^2E^2(N-1)}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\right)\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 \\ \leq & 2\frac{f(\bar{\boldsymbol{x}}_0)-\mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma}+\gamma L\sigma^2\left(\sum_{j=1}^{N}\frac{1}{n_j}\right)+\frac{6\gamma^2(N+1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\sigma^2\sum_{j=1}^{N}\frac{1}{n_j} \\ & +3N\delta_1^2+\frac{162\gamma^2N(N+1)E^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\delta_1^2 \\ & 3\frac{1-64\gamma^2E^2\left(N^2+1\right)L^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\left(\frac{1}{T}\sum_{t=0}^{T-1}T_7\right) \\ & +\frac{2(N-1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2\end{aligned} \tag{36}$$

Plugging T_7 into the above inequality, we obtain

$$\begin{aligned}&\frac{1-72\gamma^2E^2\left(N^2+1\right)L^2-162\gamma^2E^2(N-1)}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 \\ \leq & 2\frac{f(\bar{\boldsymbol{x}}_0)-\mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma}+\gamma L\sigma^2\left(\sum_{j=1}^{N}\frac{1}{n_j}\right)+\frac{6\gamma^2(N+1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\sigma^2\sum_{j=1}^{N}\frac{1}{n_j} \\ & 3\frac{1-64\gamma^2E^2\left(N^2+1\right)L^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\left(4\gamma^2NH\sigma^2+12\gamma^2NH^2\delta_2^2\right) \\ & \underbrace{+3N\delta_1^2+\frac{162\gamma^2N(N+1)E^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\delta_1^2+\frac{2(N-1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2}_{\text{Non-diminishing term}}\end{aligned} \tag{37}$$

As $\gamma \leq \frac{1}{E\sqrt{324(N-1)+72(N^2+1)L^2}}$, we have $1-\frac{162\gamma^2E^2(N-1)}{1-72\gamma^2E^2(N^2+1)L^2} \geq \frac{1}{2}$.

Hence, we have

$$
\begin{aligned}
&\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 \\
\leq& 4\frac{f(\bar{\boldsymbol{x}}_0)-\mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma}+2\gamma L\sigma^2\left(\sum_{j=1}^{N}\frac{1}{n_j}\right)+\frac{12\gamma^2(N+1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\sigma^2\sum_{j=1}^{N}\frac{1}{n_j} \\
&24\gamma^2NH\frac{1-64\gamma^2E^2\left(N^2+1\right)L^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\left(\sigma^2+3H\delta_2^2\right) \\
&\underbrace{+6N\delta_1^2+\frac{324\gamma^2N(N+1)E^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\delta_1^2+\frac{4(N-1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{w}}_{m'E}\right\|^2}_{\text{Non-diminishing term}}
\end{aligned}
\tag{38}
$$

By setting $\gamma \leq \min\left\{\frac{1}{2(N+1)E}, \frac{1}{2(N+1)E}L\right\}$, we have

$$
\frac{12\gamma^2(N+1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\leq 1,\ \frac{4(N-1)E}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\leq 8(N-1)E,\ \frac{324\gamma^2N(N+1)E^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\leq 81
$$

.

Besides, by setting $\gamma \leq \frac{1}{EL\sqrt{80(N^2+1)}}$, we have

$$
\frac{1-64\gamma^2E^2\left(N^2+1\right)L^2}{1-72\gamma^2E^2\left(N^2+1\right)L^2}\leq 2.
$$

With the above inequalities, we have

$$
\begin{aligned}
&\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_t)\right\|^2 \\
\leq& 4\frac{f(\bar{\boldsymbol{x}}_0)-\mathbb{E}[f(\bar{\boldsymbol{x}}_T)]}{\gamma}+2\gamma L\sigma^2\left(\sum_{j=1}^{N}\frac{1}{n_j}\right)+12\gamma\sigma^2\sum_{j=1}^{N}\frac{1}{n_j} \\
&48\gamma^2NH\left(\sigma^2+3H\delta_2^2\right) \\
&\underbrace{+6N\delta_1^2+81\delta_1^2+8(N-1)E\frac{1}{T}\sum_{m'=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{x}}_{m'E}\right\|^2}_{\text{Non-diminishing term}}
\end{aligned}
\tag{39}
$$

The trade-off between convergence bound and communication and computation burden

Another result:
Combining the upper bound, we have

$$\mathbb{E}[f(\bar{\boldsymbol{x}}_{t+1})] \leq f(\bar{\boldsymbol{x}}_t) - \frac{\gamma}{2}\|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 + \frac{\gamma^2 NL}{2}\sigma^2 + \frac{\gamma}{2}\left\{3N\delta^2 + 3(N-1)\|\bar{\boldsymbol{x}}_t\|^2\right\} + \frac{3\gamma}{2}\underbrace{\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left\|\bar{\boldsymbol{x}}_t^j - \boldsymbol{x}_i^t\right\|^2}_{T_7} \tag{40}$$

Telescoping the above inequality from $t=0$ to $t=T-1$ and reorganizing these terms gives

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 \leq & 2\frac{f(\boldsymbol{x}_0) - f(\bar{\boldsymbol{x}}_T)}{\gamma T} + \gamma NL\sigma^2 + 3N\delta^2 + 3(N-1)\frac{1}{T}\sum_{t=0}^{T-1}\|\bar{\boldsymbol{x}}_t\|^2 \\ & + \frac{1}{T}\sum_{t=0}^{T-1}\underbrace{\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left\|\bar{\boldsymbol{x}}_t^j - \boldsymbol{x}_i^t\right\|^2}_{T_7}\end{aligned} \tag{41}$$

Pluging the above term into (41) gives

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 \leq & 2\frac{f(\boldsymbol{x}_0) - f(\bar{\boldsymbol{x}}_T)}{\gamma T} + \gamma NL\sigma^2 + \underbrace{3N\delta^2 + 3(N-1)\frac{1}{T}\sum_{t=0}^{T-1}\|\bar{\boldsymbol{x}}_t\|^2}_{\text{non-diminishing term}} \\ & + 4\gamma^2 NH\sigma^2 + 12\gamma^2 NH^2\delta_2^2\end{aligned} \tag{42}$$

Set $\gamma = \frac{1}{\sqrt{NT}}$, we have

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(\bar{\boldsymbol{x}}_t)\|^2 \leq & \underbrace{2\frac{f(\boldsymbol{x}_0) - f(\bar{\boldsymbol{x}}_T)\sqrt{N}}{\sqrt{T}} + L\sigma^2\frac{\sqrt{N}}{\sqrt{T}}}_{\text{Dominating terms}} + \underbrace{3N\delta^2 + 3(N-1)\frac{1}{T}\sum_{t=0}^{T-1}\|\bar{\boldsymbol{x}}_t\|^2}_{\text{Non-diminishing term}} \\ & + 4\sigma^2\frac{H}{T} + 12\delta_2^2\frac{H^2}{T}\end{aligned} \tag{43}$$